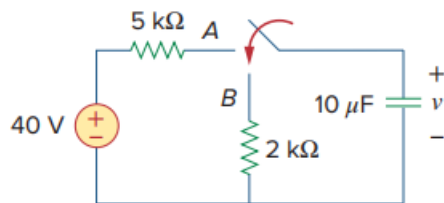


Problem

The switch in Fig. 7.84 has been in position *A* for a long time. Assume the switch moves instantaneously from *A* to *B* at $t = 0$. Find v for $t > 0$.

Fig. 7.84:



Step-by-step solution

Step 1 of 6

Consider the following circuit diagram:

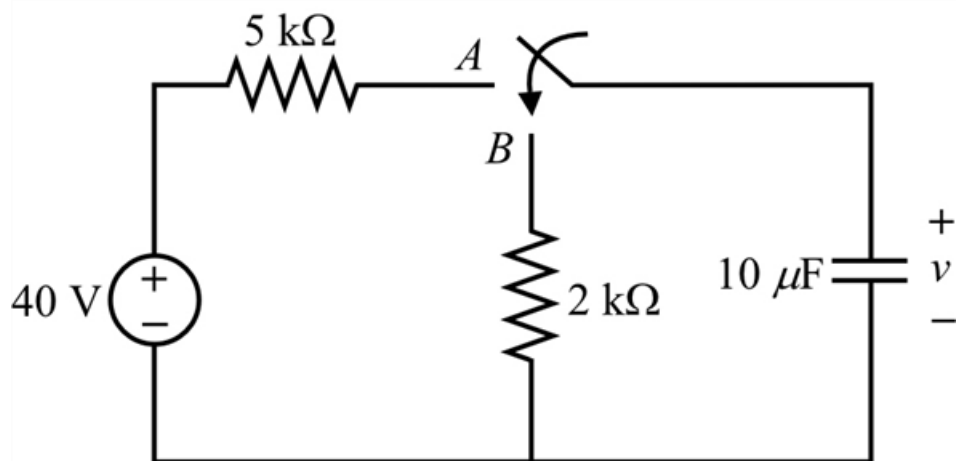


Figure 1

Step 2 of 6

For $t = 0^-$, the switch is at position A. the capacitor acts as open circuit. Draw the modified circuit diagram:

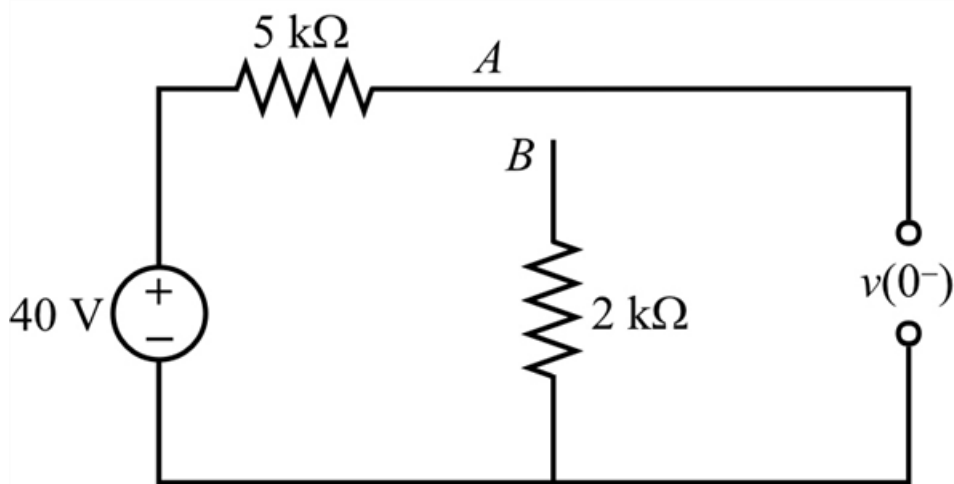


Figure 2

Step 3 of 6

Since no current flows through the circuit, so the voltage across the capacitor at $t = 0^-$ is 40 V.

$$v(0^-) = 40 \text{ V}$$

Since the voltage across the capacitor cannot change instantaneously, the voltage across the

capacitor at $t = 0^-$ is the same at $t = 0$.

$$v(0) = v(0^-) = 40 \text{ V}$$

Step 4 of 6

For $t = 0$, the switch in the circuit moves to position B. draw the modified circuit diagram.

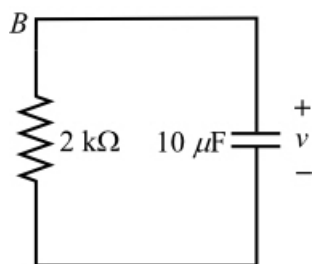


Figure 3

Step 5 of 6

Now the circuit is a source free RC circuit, write the voltage expression for RC circuit for

$t > 0$.

$$v(t) = V_0 e^{-\frac{t}{\tau}}$$

Calculate the time constant.

$$\begin{aligned} \tau &= RC \\ &= (2 \text{ k}\Omega)(10 \text{ }\mu\text{F}) \\ &= 0.02 \text{ s} \end{aligned}$$

Step 6 of 6

Substitute 40 V for V_0 and 0.02 s for τ in the voltage expression, $v(t) = V_0 e^{-\frac{t}{\tau}}$.

$$\begin{aligned} v(t) &= (40) e^{-\frac{t}{0.02}} \\ &= 40 e^{-50t} \text{ V} \end{aligned}$$

Therefore, the voltage, $v(t)$ for $t > 0$ is $\boxed{40 e^{-50t} \text{ V}}$.